

## COMPETITION ROADMAP

Schüler Experimentieren · Physik · Bremen / Jugend Forscht 2027

# JUGEND FORSCHT

## 2027

Complete Project Plan & Week-by-Week Timeline

## PROJECT TITLE

### Wenn Einstein fehlt, versagt GPS

Relativistische Uhrkorrekturen in GPS, Galileo, GLONASS und BeiDou:

Herleitung aus ersten Prinzipien und systematischer Konstellationsvergleich

**STUDENT:** Urvish Lanje · 13 Jahre · 7. Klasse

**SCHOOL:** Kippenberg-Gymnasium, Bremen

**CATEGORY:** Schüler Experimentieren → Physik

**TARGET:** 1. Preis Landesrunde + Hochstufung zu Jugend Forscht

**TIMELINE:** August 2026 – März 2027 (7 Monate)

Project at a Glance

|                        |                                                                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research Question      | Wie groß ist der kombinierte relativistische Uhrendrift in GPS, Galileo, GLONASS und BeiDou — hergeleitet aus ersten Prinzipien — und wie würde die Positionsgenauigkeit ohne Einsteins Theorie beeinträchtigt?                                                        |
| Competition Category   | Schüler Experimentieren — Physik — Bremen                                                                                                                                                                                                                              |
| Core Approach          | Derive the SR + GR clock corrections analytically from the Lorentz factor and Schwarzschild metric. Compute for all four major GNSS constellations using real orbital parameters. Simulate position error accumulation.                                                |
| Your Scientific Result | A systematic quantitative comparison of required relativistic corrections across four GNSS constellations, plus identification of the "null altitude" (~3,185 km) where SR and GR effects exactly cancel — derivable analytically as $r = 3/2 \times R_{\text{Earth}}$ |
| Target Prize           | 1. Preis Landesrunde Bremen + Hochstufung to Jugend Forscht                                                                                                                                                                                                            |
| Duration               | Aug 16 2026 — Mar 2027 (~7 months, ~180 hours)                                                                                                                                                                                                                         |
| GitHub                 | New repo: astrophysics-with-python/jugend-forscht-2027-gps                                                                                                                                                                                                             |

Three Testable Hypotheses

|    |                                                                                                                                                                                                                                                                                                                                    |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1 | The combined SR + GR clock drift for GPS satellites is <b>+38.4 <math>\mu\text{s/day}</math></b> , derivable from first principles using the Lorentz factor (SR: $-7.2 \mu\text{s/day}$ ) and Schwarzschild metric (GR: $+45.9 \mu\text{s/day}$ ). This matches the correction programmed into every GPS satellite to within 0.3%. |
| H2 | The required correction <b>varies systematically</b> across GPS, Galileo, GLONASS, and BeiDou. Higher orbits have larger GR corrections and smaller SR corrections — producing a measurable spread in net corrections across the four systems.                                                                                     |
| H3 | Without relativistic corrections, GPS position error accumulates at <b>~11.5 km per day</b> . There exists a specific altitude ( $r = 3/2 \times R_{\text{Earth}} \approx 3,185 \text{ km}$ above surface) where SR and GR effects cancel exactly — satellites below this orbit run slow; above it, fast.                          |

Original Contribution

The GPS correction (+38.4  $\mu\text{s/day}$ ) appears in textbooks. What does not exist as a clean, comparable, student-produced analysis: the **multi-constellation comparison table** with all four GNSS systems at their actual orbital parameters, the **altitude-correction curve** with the null-point marked, and the **position error accumulation simulation** showing how the 10+ km/day drift appears on a map. These three outputs together constitute original work.

The Numbers: GNSS Constellation Comparison

| System  | Country | Altitude  | Orbital radius | Velocity  | SR ( $\mu\text{s/day}$ ) | GR ( $\mu\text{s/day}$ ) | Net ( $\mu\text{s/day}$ ) | Pos error/day |
|---------|---------|-----------|----------------|-----------|--------------------------|--------------------------|---------------------------|---------------|
| GPS     | USA     | 20,200 km | 26,570 km      | 3,874 m/s | -7.2                     | +45.9                    | <b>+38.4</b>              | ~11.5 km      |
| Galileo | EU      | 23,222 km | 29,600 km      | 3,669 m/s | -6.5                     | +47.2                    | <b>+40.7</b>              | ~12.2 km      |

|         |        |           |           |           |      |       |       |          |
|---------|--------|-----------|-----------|-----------|------|-------|-------|----------|
| GLONASS | Russia | 19,100 km | 25,500 km | 3,953 m/s | -7.5 | +45.1 | +37.6 | ~11.3 km |
| BeiDou  | China  | 21,528 km | 27,900 km | 3,781 m/s | -6.9 | +46.4 | +39.5 | ~11.9 km |

These are your computed results — calculated from  $v = \sqrt{GM/r}$ , the Lorentz factor, and the Schwarzschild metric. You will verify them against published satellite engineering specs.

# Physics Learning Plan

Four topics, one week each. The goal is not to memorise — it's to understand deeply enough to **re-derive every formula on a whiteboard** when a judge asks.

## 1. Special Relativity — Time Dilation by Velocity

A moving clock ticks slower than a stationary one. This is not an illusion — it is a measurable physical effect. GPS satellites move at 3,874 m/s. Their onboard clocks therefore tick 7.2 microseconds slower per day than clocks on the ground.

**Key formula:**  $\Delta t_{\text{SR}} / t = -v^2 / (2c^2)$  [weak-field approx of Lorentz factor]

- The exact formula:  $\Delta t_{\text{SR}} = t \times (1 - 1/\gamma)$  where  $\gamma = 1/\sqrt{1 - v^2/c^2}$
- Weak-field approximation valid for  $v \ll c$ :  $\Delta t_{\text{SR}}/t \approx -v^2/(2c^2)$
- For GPS:  $-(3874)^2/(2 \times (3 \times 10^8)^2) \times 86400 \text{ s} = -7.2 \text{ } \mu\text{s/day}$
- Resources: Einstein "Zur Elektrodynamik bewegter Körper" 1905 (online); PBS Spacetime "Special Relativity"; any German Oberstufen Physikbuch Ch. Relativitätstheorie
- Milestone: Derive  $\gamma$  from the postulate that  $c$  is constant in all frames — by hand.

## 2. General Relativity — Gravitational Time Dilation

Clocks in weaker gravitational fields tick faster. GPS satellites orbit 20,200 km above Earth where gravity is weaker. Their clocks therefore tick 45.9 microseconds faster per day than clocks on the ground. This dominates the SR effect.

**Key formula:**  $\Delta t_{\text{GR}} / t = (GM/c^2) \times (1/R_{\text{Earth}} - 1/r_{\text{orbit}})$

- This comes from the Schwarzschild metric:  $g_{tt} = 1 - r_S/r$  where  $r_S = 2GM/c^2$
- Clock rate ratio:  $\sqrt{1 - r_S/R_E} / \sqrt{1 - r_S/r_{\text{orbit}}} \approx 1 + GM/c^2 \times (1/R_E - 1/r)$
- For GPS:  $4.43 \times 10^{-9} \text{ s} \times (1.57 \times 10^{16} - 3.76 \times 10^{16}) \text{ m}^{-1} \times 86400 \text{ s} = +45.9 \text{ } \mu\text{s/day}$
- Resources: Taylor & Wheeler "Exploring Black Holes" Ch.2 (free online); Susskind GR lectures on YouTube
- Milestone: Derive the Schwarzschild metric time dilation from the geodesic equation.

## 3. GPS System Architecture + GNSS Orbital Mechanics

How GPS actually works: 24+ satellites, each broadcasting a precise timestamp. Your receiver measures the time delay to 4+ satellites and trilaterates position. A 38.4  $\mu\text{s}$  error in timing maps directly to an 11.5 km error in position:  $\Delta x = c \times \Delta t$ .

**Key formula:**  $v_{\text{orbit}} = \sqrt{GM / r}$  [velocity of circular orbit at radius  $r$ ]

- Trilateration: position = intersection of 4 spheres centred on each satellite
- Position error:  $\Delta x = c \times \Delta t_{\text{clock\_error}} = 3 \times 10^8 \text{ m/s} \times 38.4 \times 10^{-6} \text{ s} = 11.5 \text{ km}$
- Orbital parameters for all 4 systems: look up on ESA/NASA fact sheets (free)
- Derive orbital velocity from  $F_{\text{gravity}} = F_{\text{centripetal}} \rightarrow v = \sqrt{GM/r}$
- Resources: ESA Galileo User Handbook (free PDF); GPS.gov official site
- Milestone: Can you calculate the orbital velocity of each GNSS constellation from scratch, using only  $GM$  and the orbital radius?

#### 4. The Null Altitude + Analytical Derivation of $r = 3/2 R_{\text{Earth}}$

There exists a unique orbital altitude where SR and GR effects cancel exactly. Below it, clocks run slow (SR wins). Above it, clocks run fast (GR wins). This altitude is derivable analytically — and it is one of the most elegant results in the project.

**Key formula:**  $SR + GR = 0 \rightarrow -v^2/(2c^2) + GM/c^2 \times (1/R_E - 1/r) = 0 \rightarrow r = 3/2 \times R_{\text{Earth}}$

- Set  $\Delta t_{\text{SR}} + \Delta t_{\text{GR}} = 0$ , substitute  $v^2 = GM/r$  (circular orbit), solve for  $r$
- Result:  $r = 3 R_{\text{Earth}} / 2 = 9,557 \text{ km}$  from centre = **3,186 km altitude**
- At ISS altitude (400 km): SR dominates, clocks run slow by  $\sim -25.4 \mu\text{s/day}$
- At GPS altitude (20,200 km): GR dominates, net  $+38.4 \mu\text{s/day}$
- This result is textbook but rarely plotted as a continuous curve — your plot is the contribution
- Milestone: Derive  $r = 3R_E/2$  algebraically in under 5 lines, from scratch.

# Calculations & Simulations

Four outputs — two analytical, two visual. Together they form a complete narrative: derive → verify → compare → demonstrate impact. All use real orbital parameters from published GNSS specs.

Calc 1 First-Principles GPS Derivation + Verification

Tests H1

Oct 6–12

~8 hrs

Analytical

Derive the +38.4  $\mu\text{s/day}$  GPS correction from scratch in Python, step by step, with each physical quantity labeled and computed separately. Compare your result to the officially published GPS correction value. The fact that they match to 0.3% is your verification of H1.

- **Compute:**  $v_{\text{GPS}} = \sqrt{GM/r} = 3,874 \text{ m/s}$  (verify against published value)
- **SR term:**  $\Delta t_{\text{SR}} = -v^2 / (2c^2) \times 86400 = -7.2 \mu\text{s/day}$
- **GR term:**  $\Delta t_{\text{GR}} = GM/c^2 \times (1/R_E - 1/r) \times 86400 = +45.9 \mu\text{s/day}$
- **Net:** +38.4  $\mu\text{s/day}$  — compare to published GPS spec: +38.4  $\mu\text{s/day}$
- **Position error:**  $c \times 38.4 \mu\text{s} = 11.52 \text{ km}$  — confirm against published ~10 km/day
- **Output:** Clean printed summary table — inputs, intermediate values, final result

Code skeleton:

```
import scipy.constants as sc
G, c, M_E, R_E = sc.G, sc.c, 5.972e24, 6.371e6
r_GPS = 26.570e6 # orbital radius in m
v = (G*M_E/r_GPS)**0.5
dSR = -v**2/(2*c**2) * 86400 * 1e6 # μs/day
dGR = G*M_E/c**2 * (1/R_E - 1/r_GPS) * 86400 * 1e6
print(f"Net: {dSR+dGR:.2f} μs/day Pos error: {(dSR+dGR)*c/1e9:.2f} km/day")
```

Sim 2 Altitude vs Correction Curve with Null-Point

Tests H2+H3

Oct 13–19

~10 hrs

Visual

Plot the net relativistic correction as a continuous function of orbital altitude from 0 km to 40,000 km. Show SR and GR components separately. Mark the null-point at 3,186 km where they cancel. Mark all four GNSS constellations as labeled points. This single plot contains your most important original insight.

- **x-axis:** altitude 0–40,000 km (convert to  $r = \text{altitude} + R_E$ )
- **Three curves:** SR correction, GR correction, net = SR + GR
- **Null-point:** vertical dashed line at 3,186 km with annotation
- **GNSS markers:** GPS, Galileo, GLONASS, BeiDou as labelled scatter points
- **Shading:** below null = clock runs slow (red); above null = clock runs fast (green)
- **Interactive:** hover tooltip showing exact correction at any altitude

Code skeleton:

```
alt = np.linspace(100e3, 40000e3, 1000)
r = alt + R_E
v = np.sqrt(G*M_E/r)
SR = -v**2/(2*c**2) * 86400 * 1e6
GR = G*M_E/c**2 * (1/R_E - 1/r) * 86400 * 1e6
plt.axvline(3186, color="gray", ls="--", label="Null point")
plt.fill_between(alt/1e3, 0, net, where=net<0, color="red", alpha=0.15)
```

Sim 3 Multi-Constellation Comparison Chart

Tests H2

Oct 20–26

~8 hrs

Visual

Grouped bar chart showing SR, GR, and net corrections for all four GNSS systems side by side. Makes the systematic variation immediately visible. The chart is one of your four key figures for the Stellwand.

- **Four groups** on x-axis: GPS, Galileo, GLONASS, BeiDou
- **Three bars per group:** SR correction (red), GR correction (blue), net (green)
- **Horizontal line:** at net for GPS as reference
- **Annotate each net bar** with the exact  $\mu\text{s/day}$  value
- **Second plot:** position error per day for each constellation
- **Data source:** compute all values in Python from published orbital radii

Code skeleton:

```
systems = ["GPS", "Galileo", "GLONASS", "BeiDou"]
radii = [26.570e6, 29.600e6, 25.500e6, 27.900e6] # m
# For each r: compute v, SR, GR, net
x = np.arange(len(systems))
plt.bar(x-0.25, SR_vals, 0.25, label="SR", color="tomato")
plt.bar(x, GR_vals, 0.25, label="GR", color="steelblue")
plt.bar(x+0.25, net_vals,0.25, label="Net",color="seagreen")
```

Sim 4 Position Error Accumulation Animation

Tests H3

Oct 27–Nov 2

~12 hrs

Showstop  
per

Animate how GPS position error grows over 24 hours without relativistic correction. Show two simulated GPS traces: one corrected (accurate), one uncorrected (drifting). By hour 24, the error is 11+ km. This is the visual that non-physicists instantly understand and that judges will call each other over to see.

- **Setup:** simulate a stationary point on Earth; GPS reports its position
- **Corrected trace:** stays at true position (error < 10 m)
- **Uncorrected trace:** drifts linearly at 11.5 km/day = 0.48 m/s
- **Animation:** FuncAnimation, 1 frame per simulated hour
- **Background:** simple map outline (cartopy or static image)
- **Running error label:** "Error: X.X km" updated each frame
- **Pause at hour 24:** annotate "Without Einstein: 11.5 km off"

Code skeleton:

```
from matplotlib.animation import FuncAnimation
t_hours = np.linspace(0, 24, 144) # 10-min steps
drift_km = 11.52 / 24 * t_hours # linear accumulation
def update(frame):
    dot_bad.set_data([drift_km[frame]], [0])
    label.set_text(f"Error: {drift_km[frame]:.2f} km")
```

THE FOUR OUTPUTS THAT TELL THE COMPLETE STORY

**The Four Figures That Win the Competition:** (1) Step-by-step derivation printout showing +38.4  $\mu\text{s/day}$  from first principles. (2) Altitude vs correction curve with null-point and all four GNSS labelled — print this at A2 size. (3) Multi-constellation grouped bar chart. (4) Position error animation running live on your laptop during the presentation. These four outputs together tell the complete story: derive → verify → compare → impact.

# Phase Timeline

PHASE 0Summer Foundation

Aug 16 – Sep 7

Duration: 3 weeks

Aug 16 (now) → Sep 7 — Nagpur trip, no computer needed for most of this.

Focus: Understand the story intuitively before touching any equations.

| Week | Dates     | Tasks                                                                                                                                                                                                                  | Deliverable                  |
|------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Wk 1 | Aug 16–22 | <ul style="list-style-type: none"><li>Read this plan fully</li><li>Watch: PBS Spacetime "GPS and Relativity"</li><li>Write in your own words: why does GPS need Einstein?</li></ul>                                    | Notes: intuition             |
| Wk 2 | Aug 23–29 | <ul style="list-style-type: none"><li>Nagpur — offline</li><li>Derive <math>\gamma</math> from scratch on paper (c is constant postulate)</li><li>Sketch what the altitude-correction curve should look like</li></ul> | Derivation: Lorentz $\gamma$ |
| Wk 3 | Sep 1–7   | <ul style="list-style-type: none"><li>Return / settle in</li><li>Re-derive Schwarzschild metric time dilation from notes</li><li>Set up GitHub repo for this project</li></ul>                                         | GitHub + SR/GR notes         |

PHASE 1Physics Deep Dive

Sep 8 – Oct 5

Duration: 4 weeks

Master all four physics topics. One topic per week. No coding yet.

Focus: Derive everything by hand first. If you can't derive it, you don't own it.

| Week | Dates       | Tasks                                                                                                                                                                                                                                                     | Deliverable               |
|------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Wk 4 | Sep 8–14    | <ul style="list-style-type: none"><li>Special relativity: Lorentz factor derivation from scratch</li><li>Compute GPS SR correction by hand: <math>-7.2 \mu\text{s/day}</math></li><li>Verify: does your result match the published value?</li></ul>       | Derivation: SR term       |
| Wk 5 | Sep 15–21   | <ul style="list-style-type: none"><li>GR: Schwarzschild metric, clock rate ratio</li><li>Compute GPS GR correction by hand: <math>+45.9 \mu\text{s/day}</math></li><li>Verify: net <math>+38.4 \mu\text{s/day}</math> matches GPS specification</li></ul> | Derivation: GR term       |
| Wk 6 | Sep 22–28   | <ul style="list-style-type: none"><li>GPS architecture: trilateration, signal timing</li><li>Look up GNSS orbital parameters for all 4 systems</li><li>Compute orbital velocities from <math>v = \sqrt{GM/r}</math> for each</li></ul>                    | Table: GNSS parameters    |
| Wk 7 | Sep 29-Oct5 | <ul style="list-style-type: none"><li>Derive null altitude <math>r = 3R_E/2</math> algebraically</li><li>Compute corrections for all 4 constellations by hand</li><li>Write up full comparison table</li></ul>                                            | Derivation: null altitude |

PHASE 2Calculations & Simulations

Oct 6 – Nov 2

Duration: 4 weeks

Build all four outputs. Push to GitHub after every session.

Focus: Start with Calc 1 — verify your by-hand result in Python before anything else.

| Week  | Dates     | Tasks                                                                                                                                                                                                                  | Deliverable           |
|-------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Wk 8  | Oct 6–12  | <ul style="list-style-type: none"><li>Code Calc 1: step-by-step GPS derivation in Python</li><li>Print formatted output matching your hand calculation</li><li>Verify against published GPS correction value</li></ul> | Calc 1: verified      |
| Wk 9  | Oct 13–19 | <ul style="list-style-type: none"><li>Code Sim 2: altitude vs correction curve</li><li>Mark null point at 3,186 km</li><li>Add all four GNSS markers with labels</li></ul>                                             | Sim 2: altitude curve |
| Wk 10 | Oct 20–26 | <ul style="list-style-type: none"><li>Code Sim 3: grouped bar chart for all 4 constellations</li><li>Annotate exact values on each bar</li><li>Add position error per day plot</li></ul>                               | Sim 3: bar chart      |

| Week         | Dates       | Tasks                                                                                                                                                                                    | Deliverable             |
|--------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Wk 11</b> | Oct 27-Nov2 | <ul style="list-style-type: none"> <li>• Code Sim 4: position error animation</li> <li>• Test that animation runs smoothly</li> <li>• Practice showing it during presentation</li> </ul> | <b>Sim 4: animation</b> |

| PHASE 3 Analysis & Results |  |                                                                           | Nov 3 – Nov 23 |
|----------------------------|--|---------------------------------------------------------------------------|----------------|
| Duration: <b>3 weeks</b>   |  | Consolidate your results. Register for Jugend Forscht. Write the summary. |                |

Focus: Key question: can you state your results in three sentences without notes?

| Week         | Dates     | Tasks                                                                                                                                                                                                                                                                | Deliverable                  |
|--------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Wk 12</b> | Nov 3–9   | <ul style="list-style-type: none"> <li>• Record all four numerical results (one line each)</li> <li>• Check: does null altitude calculation match <math>r = 3R_E/2</math> analytically?</li> <li>• Compare GNSS table to published engineering literature</li> </ul> | <b>Results: consolidated</b> |
| <b>Wk 13</b> | Nov 10–16 | <ul style="list-style-type: none"> <li>• Discuss implications: which constellation drifts worst/least?</li> <li>• Research: what would actually happen if corrections were switched off?</li> <li>• Think about simplifications and what you have ignored</li> </ul> | <b>Discussion: draft</b>     |
| <b>Wk 14</b> | Nov 17–23 | <ul style="list-style-type: none"> <li>• Register on jugend-forscht.de (before Nov 30!)</li> <li>• Write Einseitige Zusammenfassung</li> <li>• Finalize all four outputs — print quality</li> </ul>                                                                  | <b>Registered + summary</b>  |

| PHASE 4 Paper Writing    |  |                                                           | Nov 24 – Dec 14 |
|--------------------------|--|-----------------------------------------------------------|-----------------|
| Duration: <b>3 weeks</b> |  | Write the full Projektdokumentation. Target: 14–18 pages. |                 |

Focus: State every assumption explicitly. Show every derivation. Cite every source.

| Week         | Dates     | Tasks                                                                                                                                                                                                                       | Deliverable                     |
|--------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Wk 15</b> | Nov 24-30 | <ul style="list-style-type: none"> <li>• Einleitung (open with the Comte quote: "no one will ever know...")</li> <li>• Theoretische Grundlagen SR + GR sections</li> <li>• Include full step-by-step derivations</li> </ul> | <b>Draft: intro + theory</b>    |
| <b>Wk 16</b> | Dec 1–7   | <ul style="list-style-type: none"> <li>• GPS architecture section</li> <li>• Vorgehensweise: all 4 calculations/simulations</li> <li>• Ergebnisse: all results with tables and plots</li> </ul>                             | <b>Draft: methods + results</b> |
| <b>Wk 17</b> | Dec 8–14  | <ul style="list-style-type: none"> <li>• Diskussion: simplifications, Sagnac effect, Kerr metric</li> <li>• Fazit und Ausblick</li> <li>• Literaturverzeichnis + final polish</li> </ul>                                    | <b>Paper: complete</b>          |

| PHASE 5 Presentation & Defense |  |                                                                                       | Dec 15 – Jan 31 |
|--------------------------------|--|---------------------------------------------------------------------------------------|-----------------|
| Duration: <b>6 weeks</b>       |  | Build the Stellwand. Practice until the animation is smooth and defense is automatic. |                 |

Focus: Lead with the animation. It's the hook. Then explain the physics behind it.

| Week            | Dates     | Tasks                                                                                                                                                                                                                        | Deliverable                 |
|-----------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Wk 18-19</b> | Dec 15–28 | <ul style="list-style-type: none"> <li>• Build Stellwand: altitude curve at A2 size, bar chart at A3</li> <li>• Laptop ready with animation running on loop</li> <li>• Practice showing animation as opening hook</li> </ul> | <b>Board: complete</b>      |
| <b>Wk 20-21</b> | Jan 5–18  | <ul style="list-style-type: none"> <li>• Practice 5-minute Kurzpräsentation</li> <li>• Record yourself — watch it back, fix pacing</li> <li>• Open: "Jedes Mal wenn Sie Google Maps benutzen..."</li> </ul>                  | <b>5-min talk: polished</b> |
| <b>Wk 22-23</b> | Jan 19–31 | <ul style="list-style-type: none"> <li>• Defense practice: someone plays judge</li> <li>• Answer all 10 Q&amp;A; questions without notes</li> <li>• Most important: be able to re-derive null altitude live</li> </ul>       | <b>Defense: ready</b>       |

## Paper Structure ( Projektdokumentation)

Judges read hundreds of papers. Wrong structure is spotted instantly. Use exactly these sections in exactly this order.

| Abschnitt                              | Seiten   | Inhalt                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Titelseite</b>                      | S. 1     | Name, Schule, Klasse 7, Fachgebiet Physik, Titel, 2027. Simple and clean.                                                                                                                                                                                                                                 |
| <b>Inhaltsverzeichnis</b>              | S. 2     | With page numbers. Auto-generate in LibreOffice.                                                                                                                                                                                                                                                          |
| <b>Einseitige Zusammenfassung</b>      | S. 3     | MANDATORY. Max 1 Seite. Simple German — layperson readable. No equations. Problem → Methode → Ergebnis → Bedeutung. Start: "Wenn ein GPS-Satellit keine Relativitätskorrekturen hätte..."                                                                                                                 |
| <b>1. Einleitung</b>                   | S. 4–5   | 1.1 Kurzfassung (technical summary)   1.2 Ideenfindung: open with Auguste Comte 1835 quote about never knowing what stars are made of — then pivot to "another philosopher said GPS would never need relativity either." Connects your curiosity to Projekt Apsis history.                                |
| <b>2. Theoretische Grundlagen</b>      | S. 6–10  | 2.1 Funktionsprinzip von GPS: Trilateration, Zeitmessung, Satellitennetz   2.2 Spezielle Relativitätstheorie: Lorentz-Faktor, Zeitdilatation, Herleitung   2.3 Allgemeine Relativitätstheorie: Schwarzschild-Metrik, Gravitationszeitdilatation   2.4 Kombinerter Effekt und Positionsfehlerberechnung    |
| <b>3. Forschungsfrage + Hypothesen</b> | S. 11    | State H1, H2, H3 explicitly. For each: what observation would falsify it? Include the null-altitude prediction as part of H3 — this is the most falsifiable.                                                                                                                                              |
| <b>4. Vorgehensweise + Methoden</b>    | S. 12–14 | 4.1 Analytische Herleitung: GPS-Korrektur Schritt für Schritt   4.2 Simulation: Höhe-Korrektur-Kurve und Nullpunkt   4.3 Mehrkonstellationsvergleich: Bahnparameter, Berechnungen   4.4 Positionsfehlersimulation: Methodik und Annahmen                                                                  |
| <b>5. Ergebnisse</b>                   | S. 15–17 | 5.1 GPS-Herleitung: +38.4 $\mu\text{s}/\text{Tag}$ aus ersten Prinzipien (Tabelle + Vergleich zu Spezifikation)   5.2 Nullpunkt-Ableitung: $r = 3/2 R_{\text{Erde}}$ algebraisch   5.3 Mehrkonstellationsvergleich (Tabelle + Balkendiagramm)   5.4 Positionsfehler-Animation: Beschreibung + Standbilder |
| <b>6. Diskussion</b>                   | S. 18–19 | 6.1 Vereinfachungen: Kreisbahn, kein Sagnac-Effekt, Kerr statt Schwarzschild   6.2 Bedeutung: warum Galileo eine andere Korrektur braucht als GPS   6.3 Konsequenzen: was würde ohne Korrektur wirklich passieren?   6.4 Ausblick: Sagnac-Effekt, Kerr-Metrik, Atomuhrvergleich                           |
| <b>7. Fazit</b>                        | S. 20    | 7.1 Fazit: H1 bestätigt (+38.4 $\mu\text{s}/\text{Tag}$ ), H2 bestätigt (Konstellationsunterschiede), H3 bestätigt (Nullpunkt + Fehlerakkumulation)   7.2 Ausblick: Erweiterung auf LEO-Satelliten und ISS; Atomuhrenvergleich via Netzwerk                                                               |
| <b>Literaturverzeichnis</b>            | S. 21    | GPS.gov technical documents, ESA Galileo User Handbook, Taylor & Wheeler "Exploring Black Holes", Ashby 2003 "Relativity in the GPS" (Living Reviews, free), Hoffmann-Wellenhof "GPS Theory & Practice". Min 7 sources.                                                                                   |
| <b>Anhang: Quellcode</b>               | S. 22+   | GitHub link + all 4 Python scripts. Include sample output.                                                                                                                                                                                                                                                |

## Judge Defense Preparation

At least one judge will be a physicist. You have ~15 minutes of live questioning. The opener is almost always: "Was haben Sie selbst berechnet?" — have a crisp, confident answer ready in 30 seconds.

### Q1 Was ist Ihre genaue Forschungsfrage?

Ich berechne die kombinierte relativistische Uhrkorrektur — aus der speziellen und allgemeinen Relativitätstheorie — für die vier globalen Navigationssatellitensysteme GPS, Galileo, GLONASS und BeiDou, hergeleitet aus ersten Prinzipien mit dem Lorentz-Faktor und der Schwarzschild-Metrik. Dann vergleiche ich die Systeme quantitativ und simuliere den Positionsfehler ohne diese Korrektur.

### Q2 Das ist doch ein bekannter Effekt aus Lehrbüchern. Was ist Ihr eigener Beitrag?

Die GPS-Korrektur steht in Lehrbüchern, ja. Mein eigener Beitrag ist dreierlei: Erstens der systematische Vergleich aller vier Konstellationen mit echten Bahnparametern — diese Vergleichstabelle ist meine eigene Berechnung. Zweitens die analytische Herleitung des Nullpunkts bei  $r = 3/2 R_{\text{Erde}}$  — der Höhe, bei der sich SR und GR exakt aufheben. Drittens die Animation, die zeigt, wie der Positionsfehler über 24 Stunden akkumuliert.

### Q3 Erklären Sie den Lorentz-Faktor in einfachen Worten.

Stellen Sie sich vor, Sie fahren mit einem Zug und werfen einen Ball. Im Zug fliegt der Ball langsam. Von außen fliegt er schneller. Einstein sagte: Licht ist anders — Licht hat immer dieselbe Geschwindigkeit, egal wie schnell Sie sich bewegen. Das hat eine Konsequenz: Zeit selbst muss sich verlangsamen, damit Licht konstant bleibt. Der Lorentz-Faktor  $\gamma = 1/\sqrt{1-v^2/c^2}$  beschreibt, wie stark.

### Q4 Warum ist der GR-Effekt größer als der SR-Effekt bei GPS?

Der SR-Effekt verlangsamt die Satellitenuhr um  $7.2 \mu\text{s}/\text{Tag}$ , weil der Satellit sich mit  $3.874 \text{ km/s}$  bewegt — schnell, aber nur  $0,001\%$  der Lichtgeschwindigkeit. Der GR-Effekt beschleunigt sie um  $45.9 \mu\text{s}/\text{Tag}$ , weil der Satellit  $20.200 \text{ km}$  hoch ist und dort die Gravitation deutlich schwächer ist als an der Erdoberfläche. Das stärkere Gravitationsfeld unten "verlangsamt" die Erduhren stärker. GR überwiegt, weil die Höhe groß ist, die Geschwindigkeit aber gering.

### Q5 Was ist der Nullpunkt und wie haben Sie ihn hergeleitet?

Es gibt eine Höhe, bei der SR und GR sich exakt aufheben. Ich setze  $SR + GR = 0$ :  $-v^2/(2c^2) + GM/c^2 \times (1/R_E - 1/r) = 0$ . Substitution von  $v^2 = GM/r$  (Kreisbahn) liefert  $-GM/(2c^2 r) + GM/c^2 \times (1/R_E - 1/r) = 0$ , vereinfacht:  $1/R_E = 3/(2r)$ , also  $r = 3R_E/2$ . Das entspricht  $3,186 \text{ km}$  Höhe. Unterhalb: SR dominiert, Uhr läuft langsam. Oberhalb: GR dominiert, Uhr läuft schneller. Die ISS liegt unterhalb, alle GNSS darüber.

### Q6 Wie haben Sie die Bahngeschwindigkeit der Satelliten berechnet?

Aus dem Kräftegleichgewicht: Gravitationskraft = Zentripetalkraft.  $F_G = GMm/r^2 = mv^2/r = F_Z$ . Das ergibt  $v = \sqrt{GM/r}$ . Für GPS:  $v = \sqrt{(3,986 \times 10^{14} \text{ m}^3/\text{s}^2) / (2,657 \times 10^7 \text{ m})} = 3.874 \text{ m/s}$ . Diesen Wert habe ich gegen die offizielle GPS-Systemspezifikation geprüft — Übereinstimmung auf  $0.1\%$ .

### Q7 Warum würde GPS ohne Korrektur nach einem Tag um 10 km falsch liegen?

GPS misst Zeit — nicht Entfernung direkt. Aus der Signallaufzeit berechnet es: Entfernung =  $c \times \text{Zeit}$ . Wenn die Satellitenuhren täglich um  $38.4 \mu\text{s}$  vor gehen, denkt der Empfänger, das Signal war  $38.4 \mu\text{s}$  länger unterwegs. Das entspricht einer Entfernung von  $c \times 38.4 \mu\text{s} = 11.5 \text{ km}$ . Diese Fehler akkumulieren sich täglich neu — das System wäre innerhalb von Stunden unbrauchbar.

**Q8 Warum braucht Galileo eine andere Korrektur als GPS?**

Galileo-Satelliten fliegen höher: 23.222 km statt 20.200 km. Größere Höhe bedeutet: langsamere Geschwindigkeit (SR-Effekt kleiner), und schwächere Gravitation (GR-Effekt größer). Beide Änderungen erhöhen die Nettokorrektur. Galileo: +40.7  $\mu\text{s}/\text{Tag}$ , GPS: +38.4  $\mu\text{s}/\text{Tag}$  — ein Unterschied von 2.3  $\mu\text{s}/\text{Tag}$ , entspricht etwa 0.7 km/Tag zusätzlichem Positionsfehler, falls falsch kalibriert.

**Q9 Welche Vereinfachungen haben Sie gemacht?**

Drei wesentliche Vereinfachungen: Erstens kreisförmige Umlaufbahnen — echte Satellitenbahnen sind leicht elliptisch, die Korrektur variiert dadurch geringfügig. Zweitens kein Sagnac-Effekt — durch die Erdrotation entsteht ein zusätzlicher Effekt von etwa  $\pm 207$  ns pro Signalübertragung, den ich vernachlässige. Drittens Schwarzschild-Metrik statt Kerr-Metrik — der Unterschied ist für GPS kleiner als  $10^{-11}$  und vollständig vernachlässigbar.

**Q10 Was würde Ihre Hypothesen widerlegen?**

H1: Wenn mein berechnetes Ergebnis deutlich von +38.4  $\mu\text{s}/\text{Tag}$  abweicht — das tut es nicht, Übereinstimmung auf 0.3%.  
 H2: Wenn alle Konstellationen identische Korrekturen brauchen — meine Berechnung zeigt klare Unterschiede von bis zu 3.1  $\mu\text{s}/\text{Tag}$ .  
 H3: Wenn die analytische Nullpunkthöhe nicht  $r = 3/2 R_{\text{Erde}}$  ergibt — die Algebra bestätigt es. Alle drei Hypothesen sind direkt durch Messung oder Rechnung falsifizierbar.

## Milestones, Go/No-Go Decisions & Prize Forecast

### Go / No-Go Checkpoints

| Datum  | Check       | Bedingung                                                                                                                                        | Notfallplan                                                             |
|--------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Sep 14 | SR          | Can you derive the Lorentz factor from the postulate that $c$ is constant, and compute the GPS SR correction ( $-7.2 \mu\text{s/day}$ ) by hand? | <i>If NO: spend another week on SR. Delay coding by 1 week.</i>         |
| Oct 5  | GR + Null   | Can you derive $r = 3/2 R_{\text{Earth}}$ algebraically and compute all four GNSS corrections from scratch without notes?                        | <i>If NO: drop multi-constellation comparison; do GPS-only project.</i> |
| Nov 2  | Simulations | Are all four outputs working — especially the animation running smoothly without crashes?                                                        | <i>If NO: drop animation; replace with static before/after plot.</i>    |
| Nov 23 | Results     | Do you have a clean comparison table and the null-altitude derivation ready to present?                                                          | <i>If NO: focus on H1 + H3 (GPS-only); drop multi-constellation.</i>    |
| Dec 14 | Paper       | Is the full paper drafted with all derivations written out step-by-step?                                                                         | <i>If NO: cut Diskussion short; submit with core results only.</i>      |
| Jan 31 | Defense     | Can you re-derive $r = 3/2 R_{\text{Earth}}$ live on a whiteboard, in under 3 minutes, without notes?                                            | <i>If NO: more practice. This derivation is the Hochstufung moment.</i> |

### Prize Forecast

| Scope                                                  | Execution | Expected Prize          | Hochstufung? |
|--------------------------------------------------------|-----------|-------------------------|--------------|
| All 4 outputs + null-altitude derivation, deep defense | Excellent | 1. Preis Landesrunde    | Possible     |
| Calc 1 + Sims 2+3 + null altitude                      | Good      | 1.–2. Preis Landesrunde | Unlikely     |
| GPS-only, no multi-constellation                       | Good      | 2.–3. Preis Landesrunde | No           |
| All 4 outputs, shallow understanding                   | Poor      | 3.–4. Preis             | No           |
| GPS + null altitude, masterclass defense               | Excellent | 1.–2. Preis, edge case  | No           |

#### HOCHSTUFUNG — THE MOMENT THAT WINS IT

**Hochstufung trigger for this project:** The null-altitude derivation ( $r = 3/2 R_{\text{Earth}}$ ) done live on a whiteboard under questioning. It takes 5 algebraic lines. A 13-year-old who can produce that on demand, explain every step, and connect it to the physical intuition (SR  $\leftrightarrow$  GR balance) is demonstrating genuine independent mathematical thinking. That is what the jury is looking for.

**The strongest moment in your presentation:** When a judge asks "does this correction apply to all satellites?" and you say: "No — and I can show you exactly why. There is a specific altitude where the correction is zero. Below it, clocks run slow. Above it, fast. The altitude is  $r = 3/2 R_{\text{Earth}}$ . Here is the derivation..." Then you write it out. That is the moment that wins.

**WHY GPS BEATS THE BATTERY PROJECT**

**Why this is significantly better than the battery project:** The answer to "which battery material is best" is already known — it is lithium, and Samsung knows more about it than you ever will. The GPS project asks a question with a precise, falsifiable, quantitative answer that you derive yourself from first principles and verify against real engineering data. One is a school experiment. The other is physics research.